Lab 7.2: Higher-Order Functions (F#)

lab7.2.fsx

```
let lists =
    [ [ 1; 2; 3; 4; 5; 6 ]
      [ 10; -3; 7; 8; 0; 15 ] ]

lists
|> List.iteri (fun i xs ->
    printfn "=== List %d: %A ===" (i + 1) xs
    printfn "map (square)       : %A" (List.map (fun x -> x * x) xs)
    printfn "filter (even only) : %A" (List.filter (fun x -> x % 2 = 0) xs)
    printfn "fold (sum)         : %d" (List.fold (fun acc x -> acc + x) 0 xs)
    printfn "fold (product)     : %d" (List.fold (fun acc x -> acc * x) 1 xs)
    printfn "reduce (maximum)   : %d" (List.reduce (fun a b -> if a > b then a else b) xs)
    printfn "")
```